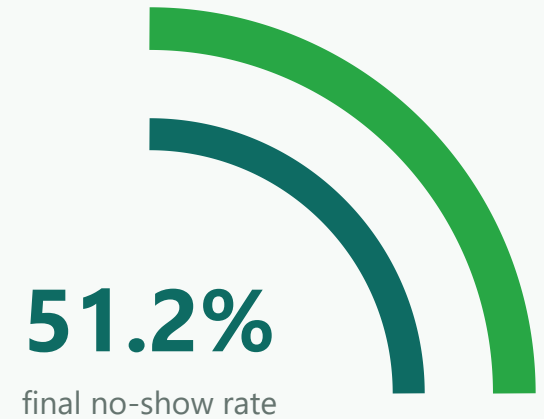


HealthConnect Appointment Analytics

Week 8 Video Presentation • Data Analytics Track

Improving patient appointment attendance
and healthcare support using data and AI.

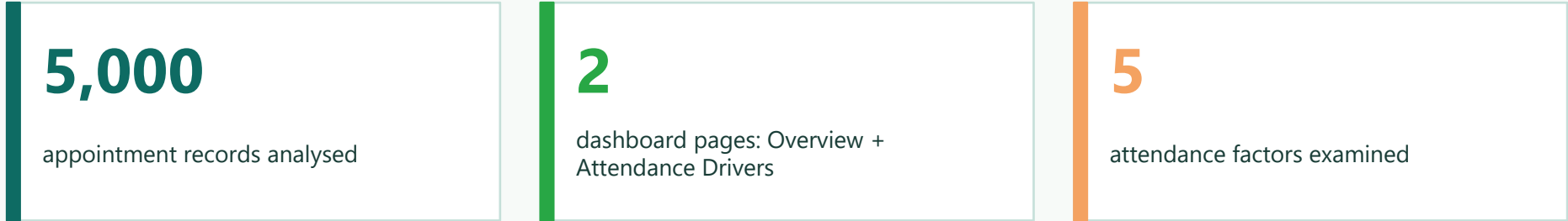
Presenter: Chenai Mpofu



My Data Analytics Contribution

Dataset, tools, dashboard structure and attendance drivers

I turned 5,000 appointment records into a Power BI decision-support dashboard.



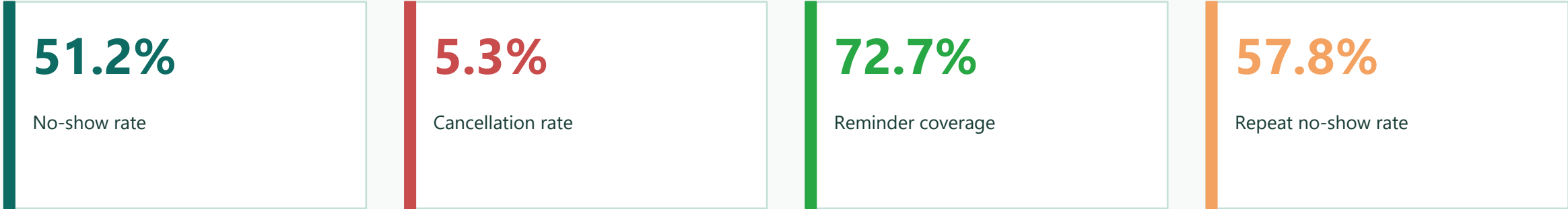
Previous no-show history
Booking lead time
Reminder status
Appointment time
Appointment type



Analysis → Dashboard → Decisions

Final Validated KPIs

Confirmed during Week 7 KPI and slicer testing



What these KPIs mean

Missed appointments are a major operational issue.
Repeat no-show behaviour deserves targeted follow-up.
Reminder coverage is an improvement opportunity.

**Final KPI set gives HealthConnect
a clear baseline for attendance
monitoring.**

Key Findings & Dashboard Insights

No-show behaviour should be explored by segment, not only as one total rate

Key insight

Missed appointments are not one single problem.

The dashboard helps decision-makers inspect attendance patterns by period and appointment type.

■ Previous no-show history	repeat-risk signal
■ Booking lead time	timing gap
■ Reminder status	process coverage
■ Appointment time	operational timing
■ Appointment type	service segmentation

Dashboard exploration

Year-Month

Appointment Type



Segmented view

Better questions:
Which month? Which appointment type?
Which driver?

Testing, Refinement & Recommendations

Validated the dashboard and translated findings into action

Test → Refine → Retest



Final recommendations

- Prioritise patients with previous no-show history.
- Improve reminder coverage.
- Monitor longer booking lead times.
- Use appointment-type segmentation.
- Review dashboard trends regularly.

Integration, Limitations & Solution Fit

Analytics connects to model interpretation and wider HealthConnect decisions

Analytics contribution

**Validated no-show
patterns + KPIs**



Data Science

**Support final model
interpretation**



HealthConnect decisions

**Targeted attendance
interventions**

Current limitations

Descriptive analysis; does not prove causality.
Reminder coverage does not prove reminder impact.
Formal cross-track evidence remains outstanding.

Broader solution

**Analytics + Data Science + ML
Engineering + GenAI + Project
Management**

Business Value & Closing

A tested analytical foundation for attendance-focused decisions

Business value

- Identifies areas needing attention.
- Monitors repeat no-show behaviour.
- Tracks reminder coverage.
- Explores patterns by month and appointment type.

Key takeaway
Analytics must be tested, interpreted, communicated clearly, and translated into action.

Future improvements

- Complete cross-track validation with Data Science.
- Investigate attendance factors more deeply.
- Compare dashboard patterns with predictive model results.

Thank you